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Prepare the new computer

ASIET Gatekeeper | Windows PowerShell setup guide | 5 October 2026

Your route

Already uploaded everything to Turso? Complete this page, then go to **Step 4**. To configure cloud access and upload more records first, follow **Steps 2 and 3** before Step 4.

1. Collect the current project

Use the latest **lib original** project, not the older Library management copy. On the current computer, finish imports and wait for every browser Scan queue to reach **0 pending**. If cloud is running in a terminal, stop it with **Ctrl+C**; for the background launcher, run **.\stop-cloud.bat** from that project folder.

Copy the project to a USB drive and name the USB folder **ASIET-Gatekeeper**. Include **.env.server**, package.json, package-lock.json, all source/configuration files, the server, src, scripts and public folders, and every .bat file. You may omit node_modules, dist, .runtime, .npm-cache and test-results. Keep your CSV exports separately as well.

2. Copy it onto the new computer

Use Windows 10/11, a local disk, an internet connection for initial setup, administrator credentials, and Edge or Chrome. Sign into the Windows account that will operate the checkpoint. Open **Windows PowerShell** from Start. These commands assume your USB is **E:**; change that letter if needed. Use a new destination folder.

```
$source = 'E:\ASIET-Gatekeeper'
$target = 'C:\ASIET-Gatekeeper'
robocopy $source $target /E /XD node_modules dist .runtime .npm-cache
if ($LASTEXITCODE -ge 8) { throw 'Copy failed. Check the paths.' }
Set-Location -LiteralPath $target
```

Robocopy exit codes 0 through 7 are normal results; 8 or higher means a copy failure. If you copied the folder with File Explorer instead, run only the Set-Location command with your actual folder path.

3. Check the required files

```
Test-Path -LiteralPath '.\setup.bat'
Test-Path -LiteralPath '.\package-lock.json'
Test-Path -LiteralPath '.\env.server'
```

All three should return **True** for migration from your existing Turso database. Never paste the contents of .env.server into a terminal or screenshot. Keep the USB and project folder private.

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Cloud setup, if needed

Skip this page if Turso already contains your completed registry and history and the copied `.env.server` is configured.

4. Install Node.js 24 for cloud development only

The local installer installs its own Node.js automatically. For running the cloud website with npm, check the system installation:

```
node --version
npm.cmd --version
```

Node must report **v24.x**. If Node is missing or has another major version, open the official Node 24 downloads below. Install the **x64.msi** package on an Intel/AMD 64-bit computer, or **arm64.msi** on ARM64. Keep npm selected. Close and reopen PowerShell afterward and repeat the checks.

```
Start-Process 'https://nodejs.org/dist/latest-v24.x/'
```

Official download source: nodejs.org/dist/latest-v24.x/. Stay on Node 24 for this project.

5. Configure the server-only environment file

```
Set-Location -LiteralPath 'C:\ASIET-Gatekeeper'
if (-not (Test-Path -LiteralPath '.\env.server')) {
    Copy-Item '.\env.server.example' '.\env.server'
}
notepad.exe .\env.server
```

If the copied file already works, leave its values unchanged. Otherwise use your existing Turso **libSQL database URL** and a **read/write database token**. Enter the following settings in Notepad; the values below are placeholders, not commands or usable credentials.

```
GATEKEEPER_DATABASE=turso
TURSO_DATABASE_URL=libsql://YOUR-DATABASE-HOST
TURSO_AUTH_TOKEN=YOUR-DATABASE-TOKEN
GATEKEEPER_GUARD_PASSWORD="YOUR-OFFICER-PASSWORD"
GATEKEEPER_ADMIN_PASSWORD="YOUR-ADMIN-PASSWORD"
```

Choose two different passwords with at least 12 characters each. Save the file as **.env.server**, not `.env.server.txt`. Keep secrets out of `VITE_*` variables and frontend files. Restart the cloud server after changing this file.

6. Install dependencies

```
npm.cmd ci
```

Wait for completion. Use `npm.cmd` in PowerShell to avoid `npm.ps1` execution-policy errors. Do not change the computer-wide execution policy.

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Start cloud and upload data

Use this stage before the one-time move to the offline computer. The website runs on this computer; its records are stored in Turso.

7. Start the cloud website - choose one method

```
.\start-cloud.bat
```

This opens **http://127.0.0.1:5173/** and keeps the API and website running in the background. You can close the launcher after it reports success. Cloud startup is manual after a reboot.

Alternatively, run the following in the project folder and **keep that PowerShell window open**. Use Ctrl+C there to stop it. Do not run both methods at once.

```
npm.cmd run dev
```

8. Sign in and import

Administrator ID: ADM-ASIET-001

Officer ID: SEC-G1-204

Cloud passwords are the values saved in .env.server.

In **Data management**, upload student and staff files, review the validation results, and complete the import. In **Administration > Import historical logs**, select the old CSV, map its columns and date format, review the preview, then import visits. Keep the same mappings when retrying an interrupted history import.

Check the member totals, staff totals, and a sample of old visits before proceeding. The migration includes members, scan events, paired visits, incidents and checkpoint settings. Browser taps that have not reached Turso are not included.

9. Finish cloud operations before handover

Stop accepting new scans and edits on **every computer** using this Turso database. Wait for all imports to finish and every Scan queue to reach 0 pending. Then stop each running cloud server.

```
.\stop-cloud.bat
```

That command stops the background launcher only. A server started with npm.cmd run dev must be stopped with Ctrl+C in its own terminal.

Next step changes where your data lives

Setup transfers and verifies Turso data, makes a verified local backup, then clears the transferred cloud records and retires that cloud dataset. This is a **one-time handover**, not ongoing synchronization. Afterward, use the installed local website only.

Keep the configured .env.server in the project on the new computer. Turso must contain your uploaded members before local setup. Both modes use API port 4317, so cloud must be stopped before local installation.

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Install the local database

Run this once on the final checkpoint computer while connected to the internet. Node.js and a separate SQL server do not need to be installed beforehand.

10. Run setup from the copied project

```
Set-Location -LiteralPath 'C:\ASIET-Gatekeeper'  
.\setup.bat
```

Start from the intended operator Windows account. Approve the Windows administrator prompt; enter administrator credentials if asked. Keep the installer open until it completes. Do not open the old cloud website during transfer.

11. Set the local login passwords

On the first installation, setup asks for an **officer password** and an **administrator password**, each with confirmation and at least 12 characters. These become the local passwords; they are not automatically copied from the cloud environment file. Store them securely.

Officer: SEC-G1-204

Administrator: ADM-ASIET-001

12. Let the installer complete these tasks

1. Download and verify a private Node.js 24 runtime.
2. Install dependencies and build the local website.
3. Create the SQLite database and protected configuration.
4. Copy the cloud registry, full history, incidents and settings.
5. Validate row counts, checksums and SQLite integrity.
6. Verify a local backup, then clear unchanged cloud records.
7. Register startup tasks and open the local website in fullscreen.

The application installs under **C:\Program Files\ASIET-Gatekeeper**. The database is **C:\ProgramData\ASIET-Gatekeeper\gatekeeper.sqlite** on a normal Windows installation. The installer uses the system Program Files and ProgramData locations.

If transfer or cleanup fails

A failed transfer stops local startup setup; fix the reported issue and rerun `.\setup.bat`. If the transfer succeeds but cloud cleanup fails, local operation can still continue and cleanup is shown as pending. Rerun setup online to retry. If cloud records changed after transfer, reconcile those records before any deletion.

13. Confirm the installation result

Look for the final **Installed. Database:** message and the website address **<http://127.0.0.1:4317>**. Sign in there using the local password you just chose. If fullscreen does not appear, continue to Step 5.

Rerunning setup preserves completed transfers and newer local records. Do not delete the local database to solve an installer error; a populated database without a transfer marker needs reconciliation.

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Verify offline use and startup

Do these checks before using the installation for live checkpoint operations.

14. Open the installed website

```
Set-Location -LiteralPath 'C:\ASIET-Gatekeeper'  
.\start.bat
```

This starts the installed server if necessary and opens the dedicated fullscreen browser profile. If you only need to inspect the page in a normal browser, its address is **http://127.0.0.1:4317**. Use the dedicated profile for normal scanning.

15. Confirm the database mode

```
$base = 'http://127.0.0.1:4317'  
Invoke-RestMethod "$base/api/config"  
Invoke-RestMethod "$base/api/system/status" |  
    ConvertTo-Json -Depth 5
```

Expect configuration **mode: local**, database engine **SQLite**, and database status **connected**. In Administration, review the cloud-transfer counts and cleanup status. Check several members and historical visits against your source exports.

16. Test without internet

Disconnect Wi-Fi/Ethernet after setup has completed. Confirm that the site still loads and member search works. With the checkpoint active, use a known card to check that a scan receives a **database-confirmed** result and its log appears. Scanning creates real records, so use an authorized test member and review the resulting IN/OUT history.

17. Test a reboot

Wait for the Scan queue to reach 0 pending, save other work, and restart Windows from the Start menu. Sign into the **same Windows account that ran setup**. The server should start at boot and the browser should open fullscreen after that account signs in. Sign into Gatekeeper again if asked.

What starts automatically

The server task runs when Windows boots. The fullscreen browser task runs when the selected Windows user signs in. Setup does **not** enable Windows automatic sign-in. Running start.bat alone does not install these tasks; setup.bat does.

If needed, open PowerShell as administrator and check that both tasks exist:

```
Get-ScheduledTask -TaskName 'ASIET Gatekeeper*' |  
    Select-Object TaskName, State
```

The tasks are named ASIET Gatekeeper Server and ASIET Gatekeeper Fullscreen. The browser task may show Ready after opening the browser. Actual startup and reboot behavior must be checked on this computer.

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Daily use, queues and backups

After handover, use the local installed website. You no longer need `npm run dev` for normal checkpoint operation.

18. Useful commands - run only the one you need

```
Set-Location -LiteralPath 'C:\ASIET-Gatekeeper'
```

```
# Start the local server and fullscreen website  
.\start.bat
```

```
# Create a backup of committed local database records  
.\backup.bat
```

```
# Stop the local server until the next start or reboot  
.\stop.bat
```

These are separate actions, not a startup sequence. Approve an administrator prompt when requested. Use **F11** to exit fullscreen or **Alt+F4** to close the browser. Closing the browser does not stop the background server.

19. Protect rapid scans and pending taps

Set the scanner to send **Enter** after each complete barcode. Complete scans enter a persistent queue and are sent in batches of up to 25. Taps leave the queue only after database confirmation. Local processing works offline.

Watch the pending count and confirmed result. If uploading stops, restore the server, sign in again if required, and use **Retry sync now**. Keep the same Windows account, dedicated browser profile and address. Do not clear browser data while taps are pending. An **unsaved-tap** warning means storage failed: keep that window open and retry after fixing storage.

This queue handles rapidly completed barcodes. Two keyboard scanners typing into the same input at exactly the same time can mix characters. A queued tap alone is not an access approval.

20. Keep backups on another disk

Run `backup.bat`, or use Administration > Local database > Create database backup. Backups contain committed records; browser-pending taps need to finish syncing or be exported separately. Backups are manual, so make them regularly.

To open the protected backup folder, use an administrator PowerShell and run:

```
Invoke-Item "$env:ProgramData\ASIET-Gatekeeper\backups"
```

Copy the new backup to a separate secure drive. A file on the same computer alone will not protect against failure of that disk. Keep recovery copies of the configuration secure too; it contains password hashes.

Clear database is not a routine migration step

Setup handles Turso cleanup. The admin Clear database control deletes members and history from the **currently connected database**. It requires a paused checkpoint, drained queues, the confirmation phrase and admin password. Local clearing creates a backup first; manual cloud clearing is permanent.

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Troubleshooting and reference

Work from the project folder. Do not delete databases, erase browser storage, or rerun imports merely to fix a connection error.

Website refused to connect / Vite connection lost

Cloud stage: run `.\start-cloud.bat` and open port 5173. For `npm.cmd` run `dev`, keep its terminal open.

Installed local stage: run `.\start.bat` and open port 4317. Cloud and local share API port 4317; do not run both simultaneously.

401 Unauthorized / login stopped working

Sign in again after a server restart or session expiry. Cloud uses passwords from `.env.server`; local uses passwords chosen during setup. An administrator login is required for administration. A 401 from a protected API is not by itself a lost database.

Missing environment file / database disconnected

Check that `.env.server` exists, has no `.txt` suffix, and contains the correct libSQL URL and database token. Cloud requires internet. For local setup, keep internet connected until downloads and transfer complete. After handover, do not try to resume the retired Turso dataset.

Setup says cloud is running / cleanup is pending

Stop the background cloud launch with `.\stop-cloud.bat`, or press `Ctrl+C` in its `npm` terminal, then rerun `.\setup.bat`. For pending cleanup, reconnect to the internet and rerun setup. If it reports changed cloud data, retain both datasets and reconcile the difference before clearing anything.

Read diagnostic logs

Cloud logs, from the project folder:

```
Get-Content '.\runtime\cloud-errors.log' -Tail 50
Get-Content '.\runtime\cloud-output.log' -Tail 50
```

Installed server log, from administrator PowerShell:

```
$log = "$env:ProgramData\ASIET-Gatekeeper\server.log"
Get-Content -LiteralPath $log -Tail 50
```

Share relevant error text only. Never share `.env.server`, tokens, passwords, member data or configuration files.

Automatic startup did not open the website

Sign into the original operator account. Check the tasks using Step 17 and inspect the local log above. Run `.\start.bat` manually. If the tasks are missing, rerun `.\setup.bat` with internet. To intentionally disable automatic startup later, run `.\uninstall-startup.bat`; it stops the server and removes tasks, but preserves data.

Scope and verification

Both addresses work only on the computer running the app. Instructions were checked against the current installer, launchers, migration/status code and package.json in lib original. Older notes saying cloud data is left intact are superseded by verified cleanup.